Antibody

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SGL 3

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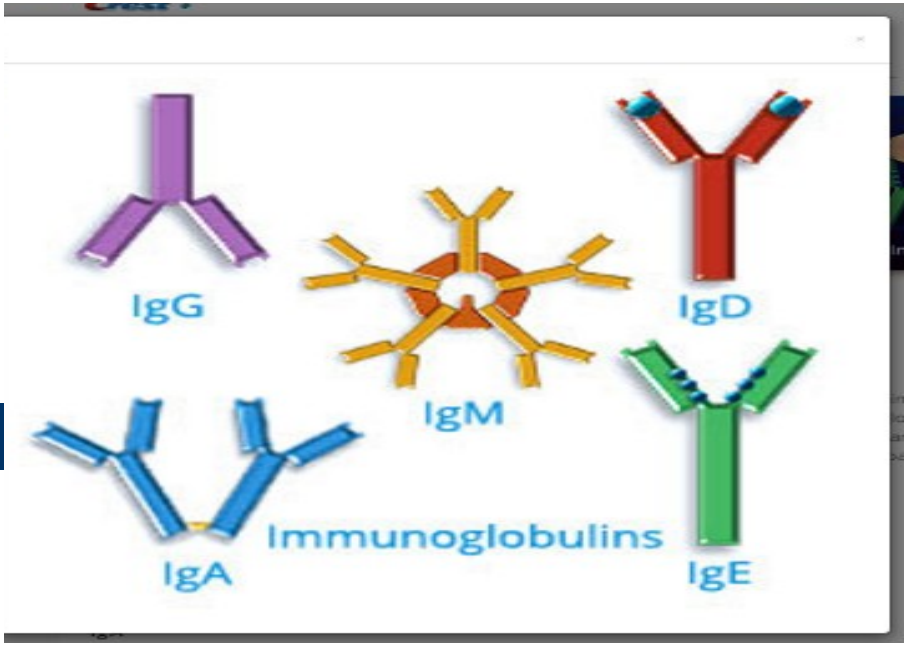
Objectives:

- The students must learn
- Definition, characteristics, classification and biological function of the antibodies.

Immunoglobulins

- **It** also known as **antibodies**, are glycoprotein molecules produced by plasma cells(WBC).
- They act as a critical part of the immune response by specifically recognizing and binding to particular antigens, and aiding in their destruction.
- Immunoglobulin occur in two main forms: soluble antibodies and membrane-bound antibodies like BcR(IgM and IgD), it is a heterodimer alpha and beta that enables the cell to transduce the signal and respond to the presence of antigens on the cell surface. The signal generated causes the growth and proliferation of the B cell and antibody production.

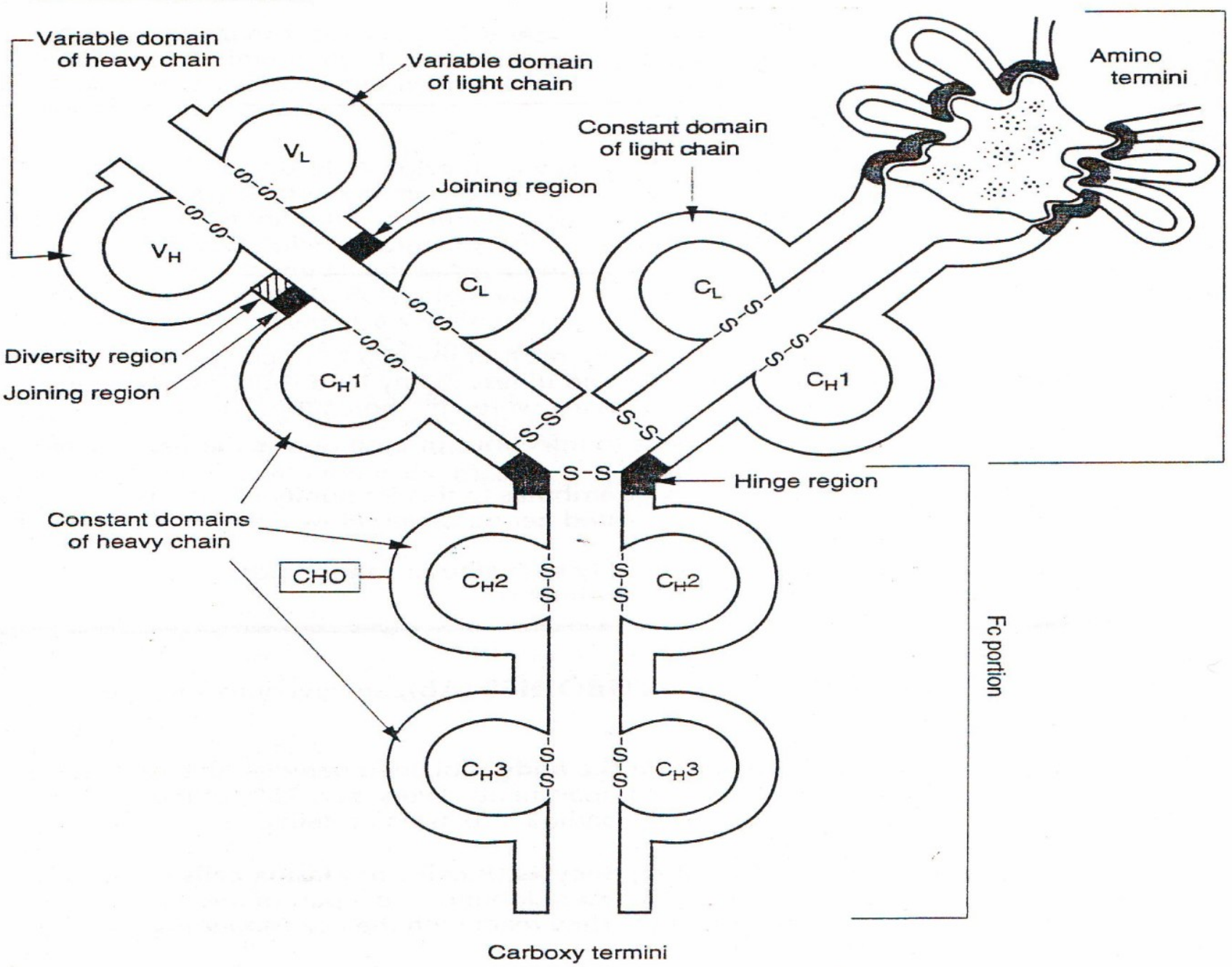
Classes of immunoglobulins



- Immunoglobulin G (IgG) [subclass IgG1, IgG2, IgG3, IgG4]
- Immunoglobulin A (IgA) [subclass IgA1, IgA2]
- Immunoglobulin M (IgM)
- Immunoglobulin D (IgD)
- Immunoglobulin E (IgE)
- Determination of individual classes and subclasses is relevant in assessing primary immunodeficiencies or immune responses.

Basic structure of immunoglobulin:-

- The stem of the Y called the **constant regions**.
- The **variable region** is antigen-binding site make up the tips of the Y's arms.
- Each chain consists of a series domains enclosed by disulfide bonds, each domain consist of 110 AA.
- Each H(heavy) chain has 4 or 5 domains one in the variable region (VH) and 3 or 4 in the constant region (CH1, CH2, CH3, and CH4).
- Each L(light)chain has 2 domains one in the variable region (VL) and one in the constant region (CL).
- The hinge region is the highly flexible portion of the H chain between the CH1 and CH2 domains, IgM and IgE do not possess a hinge.


Variable domain of heavy chain
Variable domain of light chain
Constant domain of light chain
Amino termini
Joining region
Diversity region
Joining region
Hinge region
Constant domains of heavy chain
CHO
Carboxy termini
Fab portion
Fc portion
S-S

(Hyde, 2000)

Human Immunoglobulins and their properties.					
	Immunoglobulin	Serum Level (mg/mL)	Molecular Mass (kD)	Activation/Binding of Complement	Half-life (days)
Human	IgA1	0.6–3	160 (monomer)	-	5.5
Human	IgA2	0.06–0.6	160 (monomer)	-	5.5
Human	IgM	1.5	970 (pentamer)	+++	5–10
Human	IgE	5×10^{-5}	188	-	2
Human	IgG1	3.8–11.4	146	++	23
Human	IgG2	1.5–6.9	146	+	23
Human	IgG3	0.2–1.7	165	+++	7
Human	IgG4	0.08–1.4	146	-	23

+++ = Very strong affinity; ++ = Strong affinity; + = Moderate affinity; - = No affinity

- **Immunoglobulin classes and subclasses:-**
- On the bases of AA sequences at **constant region**:
- There are 2 types of **light chain kappa κ and lambda λ**, while **heavy chains** are of **5 types**
- **Gamma γ, Alpha α, Meu μ, Delta δ and Epsilon ε.**
- **So there are 5 classes or isotypes of Igs are IgG, IgA, IgM, IgD and IgE.**
- **Igs G, D, and E are monomer structure. IgG has 4 subclasses IgG1, IgG2, IgG3 and IgG4.**

Isotype class switching

- Some B cells can undergo a “class switch” that causes the expression of a new antibody isotype. For example, the antibody isotype could switch from an IgM to an antibody of all possible classes (e.g., IgG1,...IgG4, IgE).
- During this switch, the constant region of the heavy chain is changed, but not the variable region of the heavy chain. This switch does not affect the antibody’s specificity for its antigen, but it does alter the effector functions that each class of antibody can execute.

Isotype class switching

- The antibody class switch is critically dependent on the type of cytokine that is present.
- Various cytokines, such as IL-4, IL-5, IFN-gamma and TGF-beta, are known to be responsible for class switching.

IgG and IgM

- **IgG:** It is a monomer and a major Igs in the blood, it **crosses the placenta** in the fetus.
- Play role in opsonization and involved in Antibody-dependent cell-mediated cytotoxicity (ADCC).
- Activation of complement , neutralization of toxin and antiviral activity.
- **IgM:** It is a pentamer and the first Ig produced in the immune response. Play role in agglutination, activation of complement and neutralization of toxins.

IgA: IgA exists in two isotypes, IgA1 and IgA2.

- IgA1 is monomeric and predominates in serum
- IgA2 is dimeric and a major Ig in external secretions such as saliva, sweat, gastric fluid, tears, nasal fluids, fluids of the lung and colostrum, genito-urinary tract, and gastro-intestinal tract to provides localized antibody protection on mucosal surfaces, by preventing the microorganism from adhering to, and penetrating, the mucosal epithelial lining.
- . The dimer jointed together by a J-chain and contains a secretary component or S piece by which is secreted from the epithelial cells.

Secrotary IgA biological functions:-

- Bacteriocidal activity: It has bacteriocidal activity against G-ve B.
- Antiviral activity:It prevent viruses from entering into the host cells.
- Agglutination: Secretary IgA is also an efficient agglutinating Ab.
- Complement activation: Activate the alternative pathway of the complement, it is present in colostrums and milk to protect the neonate.
- Serum IgA: In the blood, IgA interacts with an Fc receptors called **FcαRI (CD89)**, that expressed on immune effector cells, to initiate inflammatory reactions.
- Ligation of FcαRI by IgA containing immune complexes causes antibody dependent cell mediated cytotoxicity (ADCC).

IgD and IgE

- **IgD**: Present in the serum in low and variable concentration.
- It is a major surface Ig of immature B-cells and involved in their differentiation.
- **IgD biological function** in the serum is not known.
- **IgE**: Known as reagenic antibody.
- It is the **lowest** among all immunoglobulins.
- It has a unique ability of the Fc portion of IgE to bind with very **high affinity** to mast cells and basophils.
- It protects against invasion by parasites as helminthes.
- It plays a role in type I hypersensitivity.

	IgG	IgA	IgM	IgD	IgE
H-chain	γ	α	μ	δ	ϵ
H-chain subclasses	$\gamma_1, \gamma_2, \gamma_3, \gamma_4$	α_1, α_2	—	—	—
H-chain allotypes	Gm	Am	—	—	—
J chain	—	+	+	—	—
Carbohydrate (%)	3	7	12	13	11
Serum concentration (mg/dl)	1200	200	120	3	0.05
Serum half-life (days)	21†	6	10	3	2

H chain = heavy chain; J chain = joining chain.

†IgG3 has a half-life of 7 days.

(Hyde, 2000)

Immunoglobulin diversity

- All microbes can trigger an antibody response.
- Recognition and eradication of many different types of microbes require diversity among antibodies.
- Human generate about 10 billion different antibodies, each capable of binding a distinct epitope of an antigen.
- Diverse pool of antibodies from a relatively small number of antibody genes achieved by **V(D)J recombination.**

Summary:

- **Antibodies(abs)** are plasma proteins that are produced in response to immunogen with which they react specifically.
- On the bases of the immunoglobulin constant region, there are 5 Classes of Igs each with different biological functions.

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